Patrick Iben

Curriculum vitae

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GitHub portfolio | Manuscripts and teaching materials | ORCID 0009-0005-2443-7973

Research profile

Applied mathematician and biostatistician with seven years of clinical biostatistics in contract research and twelve years in research. Works to eliminate needless ambiguity, risk, and suffering in decisions that rest on data, by stating in advance how a claim could fail and reporting exactly what was measured and what was not. Develops statistical frameworks and inspectable software across clinical biometrics, AI evaluation, wearable sensing and public-data monitoring. Independent work connects model validation and reproducibility with scientific software, interactive demonstrations and educational companions. Open to full-time and consulting roles in research science, research engineering, AI evaluation, biostatistical methods and decision-support systems.

Professional experience

Biostatistician II | Syneos Health | Remote | December 2021 to October 2026

- Lead biostatistician on early-phase clinical pharmacology studies from December 2022, across first-in-human, SAD/MAD, crossover, bioequivalence and bioavailability designs.
- Owned statistical protocol sections, generated randomization schedules and provided statistical input to clinical study reports.
- Prepared statistical analysis plans and mock tables, listings and figures; created or reviewed specifications for analysis datasets and outputs using CDISC SDTM/ADaM.
- Performed PK/PD analyses and reviewed tables, figures and listings before delivery.
- Reviewed study data during trials and reviewed annotated CRFs and database designs to support planned analyses.

Associate Biostatistician | Emmes | Rockville, MD | September 2019 to December 2021

- Lead and support statistician on federally sponsored Phase 1 and Phase 3 clinical trials focused on opioid use disorder.
- Wrote protocol statistical sections, statistical analysis plans, DSMB reports and final statistical reports.
- Built dynamic progress reporting and SAS/R tools for data-management and analysis problems.
- Harmonized multi-clinic EHR data and provided statistical support to external research partners.

Research Assistant and Teaching Assistant | University of Colorado Anschutz Medical Campus | Aurora, CO | August 2016 to July 2019

- Harmonized multi-state hospital databases containing millions of records and hundreds of derived variables to estimate pediatric nursing-care costs; designed the datasets for reuse in later studies.
- Consulted on study design and sample size for doctoral nursing candidates.
- Taught graduate biostatistics review sessions and office hours.

Earlier research appointments | 2014 to 2016

- **Postsecondary and Policy Analysis Fellow, State Higher Education Executive Officers.** Analyses for the Gates Foundation-funded Multi-State Collaborative assessment of student learning outcomes.
- **Data Analysis Fellow, Yale School of Child Development.** Evaluated assistive technologies for students with print disabilities.
- **Senior Data Analyst, SCSU Office of Assessment and Planning.** Developed predictive models of student success and retention.

Education

- **MS, Biostatistics**, Data Science Analytics minor, University of Colorado Anschutz Medical Campus, 2019. GPA 3.62.

- **BS, Mathematics**, Applications of Mathematics, Southern Connecticut State University, 2016. Magna cum laude; GPA 3.78.

Clinical biometrics portfolio (synthetic data)

Clinical trial biostatistics portfolio | Source

- Built an end-to-end worked bioequivalence study on synthetic data (2x2 crossover, N = 40), from PowerTOST sample size and randomization through an ICH E9(R1) estimand and SAP, admiral ADaM, PKNCA noncompartmental analysis, the ICH M13A model, rtables/tern TLFs and CSR Section 11-style text. One script regenerates every output and CI reruns it. QC is an executed same-author re-derivation (79 NCA profiles, 0 mismatches), not independent double programming; define.xml, Pinnacle 21 and SAS are not covered.
- Implemented safety-review screening rules for a synthetic first-in-human dose-escalation study: potential Hy's Law per the 2009 FDA DILI guidance, with a 30-day window as an SAP convention; confirmation flags for ALT or AST above 3x ULN; ICH E14 QTcF categories; and a worst-of roll-up. Built a Safety Review Committee packet, a Shiny console and a 12-cell Hy's Law convention-sensitivity grid; 215 expectations in 42 tests pass. Not validated, not reviewed by a medical monitor, and author sign-off pending.
- Developed 221 ADaM-based TLF pseudocode programs per language in SAS and R for parallel, crossover, single-sequence, SAD and MAD designs. The SAS programs are not executed; the R programs are parse-checked, not run on data.
- Built an eight-domain synthetic ADaM sandbox with one-compartment oral PK/NCA and a 2-by-2 bioequivalence crossover ANOVA that regenerates its committed outputs byte for byte.
- Built a ten-gate QC harness for the portfolio. A same-author recomputation of the PK sandbox's 80 profiles by a second route agreed within a maximum relative difference of 4.4e-10. Applied established error seeding, adapted from the teeth check in the Consilience Engine software: of 18 pre-registered seeded defects, 14 were caught, 1 survived, 1 was equivalent and 2 were not applicable (counts for this list, not a detection rate). Only the recomputation gate caught the seven numeric and counting defects. Human sign-off is pending on every gate.
- Developed numeric-provenance checks for TLF interpretation (the language-model step is a stub) and custom structural dataset checks (not Pinnacle 21 or CDISC CORE).
- Ran an exploratory, ADEMP-reported residual-carryover simulation of the ICH M13A 5% pre-dose rule in a synthetic 2x2 design (n = 24). With a two-half-life washout, naive consumer risk at a true GMR of 1.25 was 0.318 versus 0.051 without residual drug; at five half-lives the rule excluded 34% of participants, unevenly by sequence. The design follows the cluster randomized trial simulation study.
- Formalized the standard C-QTc hysteresis check as an exploratory, prespecified test on the established signed loop area, with a standard error that allows for the shared time-matched placebo mean. Across 32 synthetic scenarios of 1,000 replicates its size was 0.046, while a one-sample sign-flip test over-rejected (up to 0.236) when the placebo mean was shared. Added SAD/MAD mock shells in SAS and R (SAS not executed); the framing follows the directed-contagion manuscript.
- Made three R Shiny apps (safety review, IRT/RTSM acceptance tracking and study lifecycle monitoring) run in the browser with Shinylive on bundled synthetic data, checked in Chrome only. Built an IRT/RTSM user-acceptance test library with traceability and release criteria and a coverage and continuity toolkit that runs in the browser; the SAS/R monitoring macros are not executed.

eCRF pipeline applications | Demo overview | Evaluation

- Built four Python (Streamlit) apps: Coding QC assist, De-identification review assist, Participant risk screening and Central statistical monitoring. All four also run in the browser (stlite) on bundled synthetic data with uploads disabled; the model path runs only locally.
- Separated deterministic stand-ins from optional model, Sinkhorn and Isolation Forest paths; public demonstrations and evaluations use synthetic data and placeholder dictionary codes, not MedDRA.
- Logged input hashes, configuration, paths and thresholds for traceable reruns with retained inputs and software/model artifacts; provided runnable QC.

- Published measured failure rates on synthetic held-out data: the de-identification stand-in left 108 of 308 identifiers readable (model path 91 of 308), and the coding stand-in ranked the right term first for 54.2% of verbatims (model path 77.8%). A pre-registered within-sibling check, adapted from the AlphaGenome within-site audit, routed both sibling-term families to human coding because neither family cleared the pre-registered 0.90 accuracy floor once uncertainty was counted and the tool was not shown to beat a keyword rule.

Research manuscripts

Submission and review statuses as of 23 October 2026. Public drafts, study artifacts, software and teaching materials are mapped in the research index.

Contamination-resilient cluster randomized trial design under spatial interference: a simulation study of contamination control. *Statistics in Medicine*, submitted.

- Developed a ten-diagnostic design framework and Monte Carlo comparisons using 57 public Malaria Atlas Project survey sites in Uganda. Evaluated bias, precision, interval coverage and model misspecification, distinguishing headline two-criterion simulations from full-objective cluster searches.

  Latest preprint | Study code and data | Interactive design demo | Study guide and complete companion

What do within-site comparisons of AlphaGenome Atlas predictions measure? An audit at ClinVar sites in 11 genes. *Human Genomics*, submitted.

- Audited within-site allele comparisons across 22 scorers using permutation nulls and consequence/geometry controls. Added held-out same-gene and seven-gene replications, distinguishing benchmark discrimination from clinical validity. The browser demo recomputes cached initial-cohort statistics.

  Latest preprint | Study code and data | Interactive allele demo | Study guide and complete companion

A false floor: Refuting our own information-limit claim in a lethal identification task. Submitted to PLOS ONE, 22 October 2026.

- Developed stronger-baseline counterexamples to an earlier image-information-limit claim, making yield-constrained errors and repeated-split uncertainty explicit. Added training-catalog overlap measurements and a model-coded literature audit, retaining provenance and field-safety limitations.

  Latest preprint | Provenance and training-catalog overlap records | Study guide and complete companion

Anticipating network contagions: when directedness outperforms connectedness. *Applied Network Science*, with editor.

- Developed a directional-asymmetry diagnostic and evaluated transmitter-targeted versus worst-hit control in non-negative VAR surrogates fitted to seven contagion datasets. Combined generalized connectedness, symmetrization controls and dated prediction-first records; intervention benefits were simulated.

  Author draft | Code and data | Prospective records | Study guide and complete companion

Which data modality recognizes a gene set? A domain-agnostic consilience engine. *PeerJ*, in peer review.

- Built a configurable Python engine comparing gene-group coherence across protein-domain annotations, cell-type expression, interaction associations and biological-process annotations. Applied permutation inference, multiplicity control, class-recovery analyses and method-sensitivity checks to nervous-system and immune maps.

  Author draft | Code and reproduction instructions | Study guide and complete companion

No emotion-specific image-computable code in the human left amygdala. Peer Community In, preprint review.

- Evaluated encoder-specific prediction and cross-encoder corroboration across three datasets and four analysis arms. Distinguished corrected movie effects from stronger left-amygdala session-based contrasts, with conclusions restricted to tested samples and representations.

  Latest preprint | Code and reconstruction runbooks | Study guide and complete companion

Teaching and research communication

- Authored six concise study guides and six complete educational companions tied to exact manuscript versions. All twelve editions are publicly available. The teaching sets include worked examples, answered exercises, glossaries and source-coverage maps.
- Connected manuscripts to code, data deposits, interactive demonstrations and teaching editions through a public research index. The CRT complete companion spans 97 pages; the other studies have their own full methods walkthroughs.
- Developed interactive statistical demonstrations and domain-specific explanations of the ten parameters used in response planning, supporting hover, keyboard focus and touch interaction.

Research engineering and scientific software

Public Data Consoles | Source

- Built 32 self-contained browser consoles spanning hazards, infrastructure, health-system strain, public health, access to care and markets. Examples include river-flood propagation, earthquake consequences, U.S. grid strain, NHS Scotland hospital strain, CDC surveillance and the World Bank global-health atlas.
- Integrated public observations, source-issued forecasts, published estimates and dated snapshots; implemented scheduled data publication, schema and timestamp checks, fallback behavior and source-specific freshness displays. Visitors need no provider API key.
- Adapted the ten-parameter planning framework across all 32 consoles. Response plans prioritize supported actions using the selected data, with review timing, escalation triggers, public/operator tracks and explanations of each parameter's practical relevance.
- Added mobile usability and keyboard/touch-accessible explanations; used catalog, unit and browser checks to verify source handling, plans and unavailable-feed states.

Coastal Water Resilience

- Combined CRT spatial-exposure principles and source/sink concepts informed by network-contagion research in an infrastructure planning console.
- Joined official public wastewater-station inventories to evacuation zones, USGS terrain and source-defined sewer basins, alongside NOAA/NWS observations and forecasts. Linked projected-impact screens to the same station scope used by the ten-parameter response plans.
- Incorporated Census population context and published annual treatment flows while distinguishing geographic exposure from operational outages, customer losses and hydraulic connectivity. The synthetic exercise remains a separately labeled demonstration.
- Connected parameters to phase-specific actions, explaining why a diagnostic matters at a particular timepoint and when to recheck or escalate.

Autonomic Telemetry | Source

- Built a local wearable-to-browser system using Bluetooth adapters, an in-memory Mac bridge and mobile visualization, with Work/Rest views and a fixed personal reference.
- Implemented source identity, freshness, signal-quality and channel-coverage controls, plus missing-channel handling and replayable feature exports. The physiological-pattern estimate is experimental.
- Built a separate Signal Lab with numerical lenses, analytic checks and regression tests. Live wearable input runs through the local bridge.

Resilience Telemetry | Method correspondence

- Implemented cross-asset fragility diagnostics, directed volatility networks, transmitter stability, historical replay and held-out comparisons against highest-volatility and persistence baselines.
- Integrated public market and macroeconomic data with source timestamps and interpretation limits. The ridge-regularized VAR/FEVD implementation is an exploratory adaptation of the network-contagion research; the manuscript's non-negative VAR intervention experiments are documented separately.

Consilience Engine

- Packaged gene-set analysis as an installable Python library and configurable CLI with offline cache support. Constructed modality-specific graphs from InterPro, Human Protein Atlas, STRING and Gene Ontology sources, with permutation-null separation and class-recovery analyses.

- Applied the framework to a 410-gene nervous-system map and a 132-gene immune replication. Added unit, property and metamorphic tests, clustering-sensitivity analyses, numerical tolerance checks and reproducibility manifests.

Programming and evaluation skills

- **Programming and engineering:** R (tidyverse, pharmaverse and Shiny), Quarto, SAS (Base, Macro, STAT, SQL), Python, SQL, JavaScript, LaTeX, Git, Docker, high-performance computing, public API integration and browser testing.
- **AI evaluation:** numeric grounding, provenance auditing, retrieval evaluation, model comparison, synthetic-oracle checks and reproducibility testing.

AI evaluation tools

- **numguard.** Implemented dependency-free numeric-membership, required-fact and finite phrase-denylist checks, with declarative cases, JUnit/JSON reports and CI exit codes.
- **guarded-llm-service.** Served the guard over FastAPI with evaluation and health endpoints, Prometheus-style metrics, a non-root Docker image and container smoke tests.
- **tool-grounded-agent.** Built a scripted planning prototype with tool-ledger provenance, a text guard and known-gap tests separating value membership from retrieved evidence.
- **grounded-rag.** Implemented lexical retrieval and extractive answering with retrieval metrics, numeric-membership checks and lexical-support evaluation.
- **calibrated-clinical-ml.** Implemented discrimination, calibration, recalibration, decision curves and resampling against a synthetic oracle; demonstrated threshold-decision differences between predictors with identical AUC.

Publications and acknowledged contributions

- **Acknowledged contributor:** developed data extraction and normalization programs for the study dataset in Osterhage KP, Hser YI, Mooney LJ, Sherman S, Saxon AJ, Ledgerwood M, Holtzer CC, Gehring MA, Clingan SE, Curtis ME, Baldwin LM. Identifying patients with opioid use disorder using International Classification of Diseases (ICD) codes: challenges and opportunities. *Addiction*. 2024;119(1):160-168. doi:10.1111/add.16338.
- Ben-Avie M, Rosenwald L, Depastino R, Iben P. How can we help young adults with autism thrive in the workplace? *The Conversation*, 2016.

Clinical expertise

- **Therapeutic areas:** cardiovascular, dermatology, immunology, neurology (MS, ALS, Parkinson's, Alzheimer's), oncology (hematologic malignancies and solid tumors), psychiatry (opioid use disorder and schizophrenia), and respiratory.
- **Study designs:** first-in-human, SAD/MAD, crossover, fixed sequence, parallel group, adaptive, placebo-controlled, double-blind, open-label and cohort.
- **Statistical methods:** PK/PD and NCA, bioequivalence, survival and longitudinal analysis, missing-data imputation, simulation, permutation and resampling inference, multiplicity, cluster randomized designs, calibration and decision curves.
- **Clinical standards and tools:** CDISC SDTM/ADaM, MedDRA and WHODrug coding, Veeva Vault, TLF specifications and review.